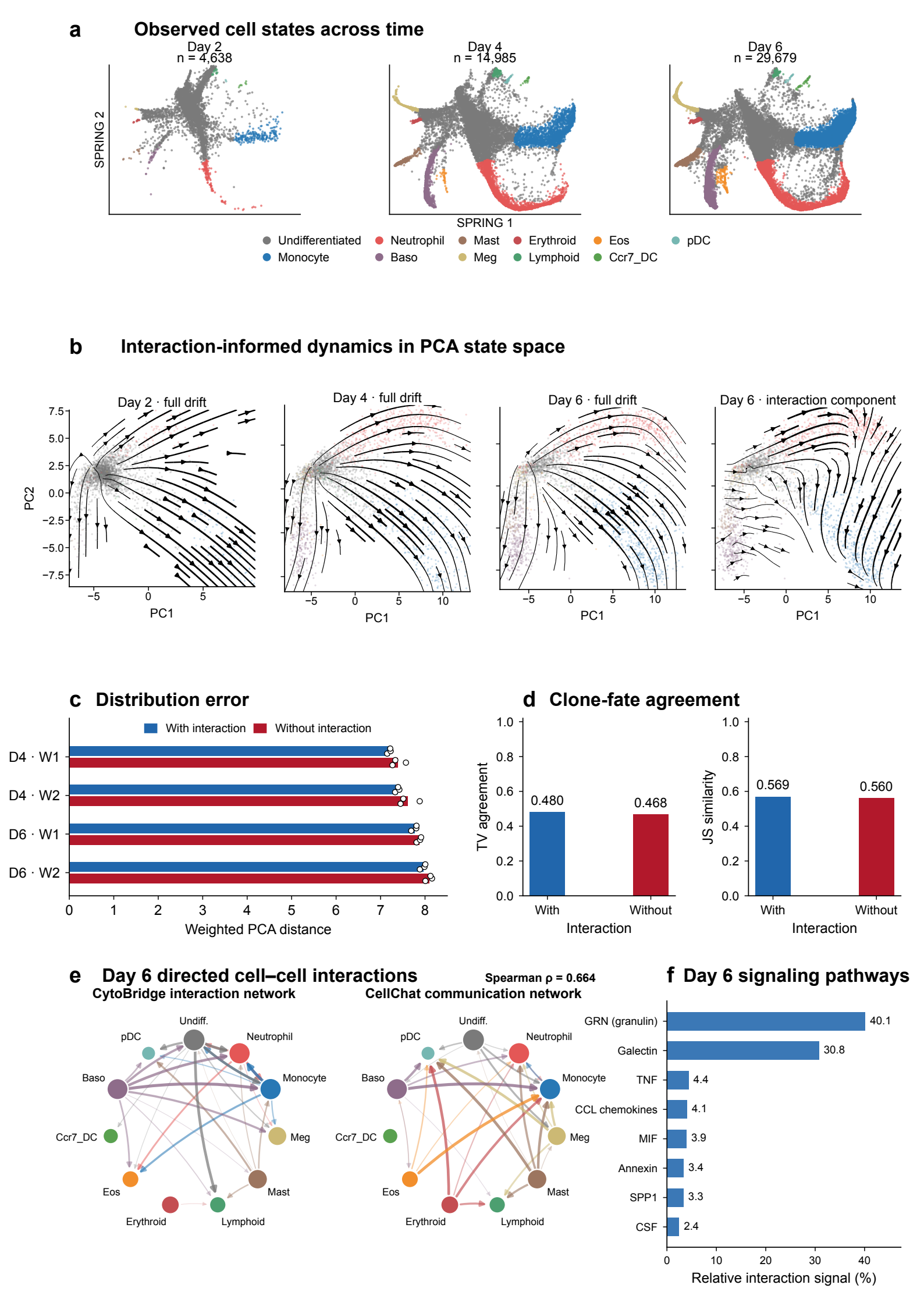

a
Observed cell states across time
Day 2
n = 4,638
Day 4
n = 14,985
Day 6
n = 29,679
SPRING 2
SPRING 1
Undifferentiated
Monocyte
Neutrophil
Baso
Mast
Meg
Erythroid
Lymphoid
Eos
Ccr7_DC
pDC
b
Interaction-informed dynamics in PCA state space
Day 2 · full drift
Day 4 · full drift
Day 6 · full drift
Day 6 · interaction component
PC1
PC2
c
Distribution error
With interaction
Without interaction
D4 · W1
D4 · W2
D6 · W1
D6 · W2
Weighted PCA distance
d
Clone-fate agreement
TV agreement
JS similarity
0.480
0.468
0.569
0.560
With
Without
Interaction
e
Day 6 directed cell–cell interactions
Spearman ρ = 0.664
CytoBridge interaction network
CellChat communication network
Undiff.
pDC
Neutrophil
Baso
Monocyte
Ccr7_DC
Meg
Eos
Mast
Erythroid
Lymphoid
f
Day 6 signaling pathways
GRN (granulin)
Galectin
TNF
CCL chemokines
MIF
Annexin
SPP1
CSF
40.1
30.8
4.4
4.1
3.9
3.4
3.3
2.4
Relative interaction signal (%)